

UNIT 4

Data preprocessing is the essential process of preparing raw data for analysis and model building by addressing quality issues and transforming it into a more efficient format.

Data Cleaning: Handling Missing Data

Missing data occurs when no value is stored for an observation.

Techniques for handling it depend on the nature of the missingness:

- **Deletion:**
 - **Row Deletion:** Removing the entire record if any value is missing. This is best for small amounts of missing data (typically <5%).
 - **Column Deletion:** Dropping an entire feature if it has a high percentage of missing values (e.g., >60–85%).
- **Imputation:**
 - **Statistical Measures:** Replacing missing values with the **Mean** (for normally distributed data), **Median** (for skewed data), or **Mode** (for categorical data).
 - **Time-Series Techniques:** Forward Fill (ffill) and Backward Fill (bfill) are foundational imputation techniques used to handle missing data in time-series analysis. They ensure that time-ordered data maintains its sequential structure by filling gaps using neighboring valid observations.

1. Forward Fill (Propagating Last Valid Value)

Mechanism: If x_t is missing, it is replaced with x_{t-1} .

2. Backward Fill (Using Next Valid Value)

Mechanism: If x_t is missing, it is replaced with x_{t+1} .

Data Cleaning: Handling Outliers

Outliers are data points that deviate significantly from the rest of the observations.

Detection Methods:

Z-Score: A z-score (standard score) measures how many standard deviations a data point is above or below the mean, calculated as

$$z = \frac{(x - \mu)}{\sigma}.$$

Interquartile Range (IQR): The interquartile range (IQR) measures statistical dispersion by calculating the difference between the upper quartile (Q3 or 75th percentile) and the lower quartile (Q1 or 25th percentile).

Formula: $IQR = Q_3 - Q_1$.

Visualization: Using **Box Plots** or **Scatter Plots** to visually identify extreme values.

- **Handling Techniques:**

Trimming: Removing the outlier rows from the dataset.

Capping (Winsorization): Replacing outliers with the maximum or minimum "normal" value

Given data: [10, 12, 14, 15, 16, 18, 100]

the value 100 is an extreme outlier.

Winsorized data: [10, 12, 14, 15, 16, 18, 18]

(100 is capped at 18).

How it Works:

Define Percentiles: Determine lower and upper percentiles

Identify Outliers: Any data point below the lower percentile or above the upper percentile is flagged.

Cap the Values: Values lower than the minimum bound are replaced with the minimum threshold value, and values higher than the maximum bound are replaced with the maximum threshold value.

Transformation: Applying logarithmic or square root transformations is a standard data preprocessing technique used to reduce right-skewness (positive skew) in datasets, making them more symmetric and normally distributed.

These transformations compress extreme high values in the "tail" of the distribution, decreasing the impact of outliers and stabilizing variance.

Data Transformation Techniques

Transformation converts data into formats better suited for algorithms.

- **Feature Extraction (PCA):**

Principal Component Analysis (PCA) is an unsupervised technique that reduces dimensionality by transforming correlated features into a smaller set of uncorrelated "principal components" while preserving maximum variance.

PCA:

- **Uncorrelated Components:** PCA transforms original variables into uncorrelated components using Eigenvectors of the covariance matrix.
- **Variance Maximization:** The first principal component holds the highest variance.
- **Data Scaling Importance:** PCA is sensitive to variance; data must be standardized (mean=0, variance=1) before analysis to avoid dominance by large-scale features.
- **Dimensionality Reduction:** It lowers the number of features, which helps in cleaning data, improving machine learning model performance, and accelerating training times.

Common Applications:

- **Data Visualization:** Reducing complex datasets to 2D or 3D for plotting.
- **Data Compression:** Representing large datasets with fewer variables.

- **Feature Engineering:** Reducing noise in predictive models.
- **Feature Selection:** Unlike extraction, selection involves choosing a subset of the original features based on statistical importance or model performance (e.g., using Filter methods).
- **Encoding Categorical Data:**
 - **Label Encoding:** Label encoding is a data preprocessing technique in machine learning that converts categorical text data into numerical integers (0 to n-1).
It is essential for algorithms requiring numerical input.
 - **One-Hot Encoding:** One-hot encoding is a preprocessing technique that converts categorical data into a numerical binary format (0s and 1s) suitable for machine learning algorithms.

It creates new binary columns for each unique category, marking "1" for the presence of a category and "0" otherwise.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a critical approach in data science used to **analyze and summarize datasets**, often employing visual methods to understand data characteristics, spot anomalies, and test hypotheses before formal modeling. EDA is generally divided into three main types based on the number of variables being analyzed: Univariate, Bivariate, and Multivariate analysis.

1. Univariate Analysis

Univariate analysis focuses on a single variable (one "variate") at a time. It is the simplest form of analysis, designed to describe the data, identify patterns, and detect outliers.

2. Bivariate Analysis

Bivariate analysis examines the relationship between two variables to determine if an empirical connection or association exists

3. Multivariate Analysis

Multivariate analysis involves investigating three or more variables simultaneously to understand complex interrelationships.

Summary Table of EDA Types

Feature	Univariate	Bivariate	Multivariate
Variables	1	2	3+
Goal	Describe Distribution	Find Relationships	Identify Interactions
Plots	Histogram, Box plot	Scatter plot, Heatmap	Pairplot, 3D Plot, Cluster Map

Statistics	Mean, Std Deviation	Correlation, Covariance	PCA, Factor Analysis
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Data visualization

Data visualization is the **graphical representation** of information, using **visual elements** like charts, graphs, and maps to make **complex data sets** accessible, **understandable**, and actionable.

It bridges the gap between **raw data analysis** and **human understanding**, accelerating decision-making by revealing **hidden patterns, trends, and outliers**, particularly in large datasets.

Need for Data Visualization

Rapid Pattern Recognition: Humans process visuals faster than spreadsheets, making it easier to spot trends and relationships.

Complex Data Simplification: It turns large, complex, and raw numerical data into easily digestible visual stories.

Data Storytelling & Communication: Effectively communicates insights to stakeholders and highlights key findings.

Identifying Outliers: Facilitates the immediate recognition of errors or anomalies within datasets.

Shapes of Data (Data Types)

Data visualization generally deals with two main types of data:

- **Numerical Data (Quantitative):** Data representing quantities and measurements, such as height, weight, temperature, or financial figures. This includes **continuous data** (e.g., measurements) and **discrete data** (e.g., counts).
- **Categorical Data (Qualitative):** Data representing groups or attributes, such as names, gender, rankings, or types of products.

Input for Data Visualization

Effective visualization requires structured data, which often includes:

- **Cleaned/Processed Datasets:** Raw data must be cleaned, transformed, and organized into rows and columns before visualization.
- **Key Variables/Metrics:** Identification of independent and dependent variables, quantitative measurements, and qualitative categories.
- **Data Sources:** Data can come from databases, spreadsheets (CSV, Excel), APIs, or real-time streaming data.

Conscious vs. Unconscious Thought

Conscious: You are aware of the thought and its source.
Example: Actively trying to remember a password.

Thought processes that happen outside of your direct attention, often in the form of impulses, memories, or

unexpected ideas. Example: A forgotten memory suddenly appearing.

Types of Data Visualization: Cognitive and Perceptual

Data visualization techniques often leverage both cognitive and perceptual abilities to be effective:

- **Perceptual Visualization (Pre-attentive Processing):** Focuses on maximizing/utilizing/ the human brain's ability to instantly recognize visual patterns *before* conscious thought (active voice).
 - *Examples:* Color, size, position, shape, and orientation. A bright red line on a chart immediately signals a sharp increase or anomaly.
- **Cognitive Visualization (High-level Processing):** Requires conscious thought and analysis to interpret the information, focusing on understanding relationships and patterns.
 - *Examples:* Complex charts like heat maps, scatter plot matrices, or 3D visuals that require interpreting multiple dimensions of data simultaneously.

Perceptual Data Visualization (Visual Input & Perception)

Perceptual techniques rely on fast, subconscious visual processing to distinguish data.

Gestalt Principles: Grouping data by proximity, similarity (color/shape), or enclosure to help the brain identify patterns.

Pre-attentive Attributes: Using visual cues that the brain processes instantly—such as length (bar charts), position (scatter plots), and color intensity (heat maps)—to highlight trends without conscious effort.

Visual Dimensions: Utilizing 3D, size, and orientation to encode multiple variables simultaneously.

Cognitive Data Visualization (Information Processing)

Cognitive techniques focus on how the brain interprets, understands, and remembers visual information.

Reducing Cognitive Load: Minimizing extraneous information to help the brain process data faster and reduce misinterpretation.

Exploratory Visualization: Designing tools that allow users to interact with data to build knowledge, understand trends, and form hypotheses.

Explanatory Visualization: Telling a story or conveying a specific message to an audience, guiding them to a clear, actionable conclusion.

Narrative Analytics/Storytelling: Contextualizing data to make it more memorable and persuasive.

Common Visualization Methods

- **Charts & Graphs:** Bar, line, pie, scatter plots, and histograms (for comparing trends, correlations, and distributions).

- **Maps:** Heat maps and geographical maps (for spatial data).
- **Dashboards:** Interconnected visuals for real-time monitoring.

Practicing Good Ethics in Data Visualization

- **Accuracy & Honesty:** Represent data truthfully without manipulating axes (e.g., not starting a bar chart at zero)
- **Contextualization:** Include necessary context to prevent misinterpretation, as skewed data can bias perception.
- **Transparency:** Clearly communicate data sources, limitations, and any potential bias or omitted data.
- **Accessibility & Privacy:** Ensure visualizations are accessible to users and protect sensitive, personally identifiable information.

Principles of Visual Perception (Gestalt Principles)

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- **Proximity (Law of Nearness):** Objects that are close to one another are perceived as a group, even if they differ in shape or size. Proximity helps organize information and create relationships without borders.
- **Similarity:** Elements that share similar characteristics—such as shape, color, size, or texture—are perceived as belonging together, forming a group or pattern.

- **Closure:** The human brain tends to fill in gaps in an incomplete image or shape to perceive a complete, whole object. It enables us to interpret incomplete information as a finished, recognizable pattern.
- **Continuity (Law of Good Continuation):** The eye tends to follow the smoothest, least interrupted path when looking at lines or curves, perceiving them as related rather than separate parts.
- **Figure-Ground:** The eye differentiates an object (the figure) from its surrounding area (the ground or background).
- **Symmetry and Order:** The brain tends to perceive objects as symmetrical shapes that form around a center. This principle suggests that we prefer balanced, orderly, and simple compositions.
- **Common Fate (Common Region):** Elements that appear to move or point in the same direction are perceived as more related than those moving in different directions.
- **Simplification:** Reduce clutter (disorder) to reduce cognitive (mental) load, allowing viewers to grasp the key message instantly.
- **Pre-attentive Attributes:** Use visual cues like color, length, and position to instantly communicate differences before conscious (sensible) attention.
- **Color Usage:** Use color carefully to highlight, not decorate.

- **Power BI:**
- **Best for:** Organizations using Microsoft 365/Azure, users needing quick setup and ease of use.
- **Cost:** Highly cost-effective; free Desktop version
- **Key Features:** Strong DAX (Data Analysis Expressions) for complex calculations, AI-powered insights, and extensive built-in connectors.
- **Tableau:**
- **Best for:** Deep, ad-hoc data exploration, sophisticated visualizations, and advanced analytics.
- **Cost:** Higher investment
- **Key Features:** Superior visual flexibility, handles massive, dissimilar datasets efficiently, strong community, and Tableau Prep for data cleaning.
- **Tableau vs Power BI Comparison:**

	Power BI	Tableau
Parent Company	Microsoft	Salesforce
Establishment Year	2013	2003
Pricing	Affordable pricing with Premium and Pro plans.	Higher cost with traditional licensing models
Licensing	No licensing is required	Licensing is required
Usability	User-friendly interface, ideal for beginners and business users	Designed for advanced users with a steeper learning curve

Scalability	Scalable for businesses of all sizes, particularly small to medium enterprises	Highly scalable for large organizations and complex projects
Database Connectivity	Best for Microsoft databases and platforms, with broader cloud compatibility.	Connects to diverse databases, including on-premises and cloud sources
Learning Curve	Easier to learn and adopt, especially for users familiar with Microsoft products	Requires more time and expertise due to its advanced features
Functions	Rely on DAX (Data Analysis Expressions) and M functions for calculations	Uses Level of Detail (LOD) expressions for calculations
Dimensions and Measures	Uses DAX for creating calculated columns and measures	Uses MDX for managing dimensions and measures
Data Preparation	Built-in Power Query simplifies data cleaning and transformation tasks	Requires Tableau Prep for complex data preparation
Deployment Speed	Faster deployment, especially for organizations using Microsoft tools	Slightly slower due to advanced setup and configuration options
Supports SQL	Yes	Yes
Data Volume	Best for small to medium-sized datasets; performance may drop with very large data	Handles large datasets efficiently, suitable for enterprise-level analysis
Advanced Analytics	Integrates with Azure AI for machine learning and natural language processing.	Includes powerful tools for predictive modeling and statistical analysis.
Trial Period	60-day free trial for Pro and Premium versions.	14-day free trial
Easy to use	Easy to learn and use	It takes time and expertise to master

